Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2414)

Key

SECTION – I

1. C
2. D
3. B
4. D
5. C
6. D
7. D
8. C
9. B
10. D
11. A
12. D
13. C
14. B
15. C
16. D
17. A
18. A
19. 2143
20. C
21. 3214
22. C
23. 2314
24. B

SECTION – II

1. 21
2. B
3. A
4. 15
5. B
6. 900
7. 10
8. D
9. B
10. B
11. C
12. 5
13. A
14. C
15. A
16. C
17. D
18. A
19. B
20. C

SECTION – III

1. 22
2. 2
3. C
4. B
5. D
6. A
7. A
8. B
9. 16
10. D
11. 900
12. 1
13. 5
14. C
15. 9
16. B
17. C
18. D
19. A
20. C
21. 27.5625
22. B

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 523

1. Consider this: ‘*Critical human geographers have adopted a more political definition of culture… Rather than viewing culture as a fixed attribute of societies, this approach views culture as always* ***emerging out of struggles among groups with diverging interests and unequal access to power.***’ If it’s a presupposition it must be true for this definition to be true.

Option A: Culture emerges out of the struggles between various groups with unequal power. We are not sure if cultures (as opposed to a singular culture) exist independently and are constantly changing or are dynamic. Option A is not the answer.

Option B: The discussion is about culture emerging out of the clash of groups. It has got nothing to do with multiple cultures existing together or with cultures clashing against each other. Option B is not the answer.

Option C: Culture emerges out of struggles between groups with different levels of access to power. Hence, we can say that culture is a function of social struggle for power. Option C is the answer.

Option D: The presupposition/assumption that stronger cultures may fail to undermine weaker ones is not necessary to argue that culture emerges out of the clash between groups. Further, the passage mentions that ‘*dominant values and practices are maintained through the cooption and dilution of alternative and often oppositional ways of life*’. From this we cannot infer that stronger cultures can fail against weaker cultures. Option D is not the answer. Choice (C)

2. Consider the statement: ‘***The denigration of the Appalachian region and subculture in popular film***—*for example, the perpetuation of the “backward” hillbilly stereotype in Sergeant York and Deliverance*—***helps urban and suburban***

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Americans to feel better about their own material conditions and lifestyle choices.'

Option A: Popular film from this particular example didn't challenge the hegemonic representations (the popular or majority view). It perpetuated it by further denigrating/belittling the subcultures. Option A is not the answer.

Option B: The perpetuation of the hillbilly stereotype didn't challenge the existing representations, it further perpetuated it, to make the hegemonic communities feel better. Option B is not the answer.

Option C: The hillbilly stereotype was not about urban and suburban Americans but about the subcultures of the Appalachian region. So, Option C is not the answer.

Option D: The author will approve of this statement, since the hillbilly stereotype misrepresented the Appalachian region, and secondly, it did so in order to perpetuate the hegemonic representation about the superiority of urban and suburban lifestyles. Option D is the answer. Choice (D)

3. Consider the statements: '***Some geographers focus on what they perceive as misrepresentations of places*** *and seek to correct errors found in news stories or even fictional accounts.* ***Other geographers****, reacting to critiques of the ability of researchers to objectively know and portray their research subjects that appeared across the social sciences during the 1980s and 1990s,* ***theorize that mass-media images contribute to, or are actually parts of, the production of social spaces and places.***'

   Option A: While it can be understood from the statements above that there are two sets of geographers who believe in two different things about mass-media images, and so, they are indeed divided in their opinion, geographers do not really talk about 'mass media destroying cultural spaces'. Option A is not the answer.

   Option B: This can be inferred from the second bold part above, that mass media images contribute to the production/creation of social spaces. In other words, because of mass media a particular place can take up media significance and end up being a social space. Option B is the answer.

   Option C: It cannot be inferred that the first set of geographers, who seek corrections of misrepresentations of places and cultures are necessarily ignoring tourists or the benefits of tourism. The two are unrelated aspects, where one set of geographers are out to correct the misrepresentations while another set believes in the perpetuation of tourism. Option C is not the answer.

   Option D: Mass media doesn't recreate social space in the mind of tourists. It is the other way *may be altered or preserved for years so that the landscape continues to recall the film in the minds of tourists.*' Option D is not the answer.

   Choice (B)

4. Option A: From '*Some of this work follows older traditions in cultural geography and maps the hearths and diffusion of particular musical forms,*' we can infer that human geographers study/understand the diffusion of the traditional forms of music. Option A is not the answer.

   Option B: The passage says that '*Other research focuses on how music lyrics establish or reproduce particular senses of place.*' This explains Option B about the connection between places and lyrics. Option B is not the answer.

   Option C: This can be understood from '*John Lovering's study of the global music industry provided insights into* ***how the processes of commodification and globalization interact with local music*** *generation and performance.*' Option C is not the answer.

   Option D: According to the passage, '*some of this work follows older traditions in cultural geography and maps the hearths and diffusion of particular musical forms.*' The work follows older traditions to study the diffusion of musical forms. This is the methodology used in the work/study. The study itself does not provide an understanding of how to use older traditions of cultural geography to map the spread of music. Rather, it makes use of the older traditions. Hence, this is the answer.

   Choice (D)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 533

5. According to Polybius' anacyclosis, the cycle is from barbarism to monarchy to aristocracy to democracy to barbarism. Three of the options are inconsistent with this cycle, while one is consistent with it, which is the answer to the question.

   Option A: According to Polybius the cycle involves aristocracy replacing kings when they are not strong enough. The political change in Option A therefore is inconsistent with the argument, by stating that one strong ruler replaces another. Option A is not the answer.

   Option B: According to Polybius, democracies decline and end up being mob rules, eventually going back to being a monarchy. Option B goes against that. Option B is not the answer.

   Option C: This is in sync with Polybius' argument where monarchs are overthrown by other subjects who would then establish an aristocracy. Option C is the answer.

   Option D: Aristocracies don't revert to monarchies according to Polybius' theory. Hence, Option D is around, that the social spaces recreate what the tourists had already seen on mass media as understood from: '*In fact, entire neighborhoods*

not the answer.

Choice (C)

6. The reference made is to: '*Polybius takes an even larger view, treating the development of humankind from its savage beginnings to civilized states and societies, even attempting to integrate this with his discussion of political change…*' So, the larger view is that of how humankind evolved from savagery to civilization.
Option A: Polybius' opinion was not restricted to Greece alone, at least in the passage. Hence, Option A is not the answer.
Option B: The cycle mentioned in this option is part of the discussion of the second change mentioned in the passage. In the first para, the larger view is about how Polybius expanded the scope of study from one particular civilization to entire humankind. The first para deals with the '*development of society from early times to their own day*' and not about the cycle of political change. Option B is not the answer.
Option C: This is not the larger view about political change that Polybius concerns himself with. It has been mentioned as the second area of interest of historians. The larger view is with respect to how Polybius took the study to the next level, by expanding the scope to humanity as a whole, and not just one kingdom (like Greece, in the case of Thucydides). Option C is not the answer.
Option D: Polybius' larger view is about the cycle of humanity from barbarism to civilization. The larger view is treating the development of humankind from its savage beginnings to civilized states and societies (it is larger because the previous view mentioned was only about Greece). He also attempts to include this in his discussion of political change, which is discussed in the subsequent paras. Hence, Option D is the answer.
Choice (D)

That is precisely what Sallust and Livy also opined. Hence, Option D is the answer. Choice (D)

8. Option A: This was indeed one of Polybius' argument as understood from '*but his descendants live luxuriously and wantonly, and this causes the best men to overthrow the monarchy…*' Option A is not the answer.
Option B: Consider this: '*Over time the people in their turn become corrupted and descend into such savagery that their only hope is a monarch.*' The discussion also involves moral decline in the last paragraph which is considered one of the causes of civil wars. Option B is not the answer.
Option C: Anacyclosis starts with barbarism and not monarchy. A strong ruler controls such a crowd and ushers in a new phase. Option C is the answer.
Option D: This can be understood from the passage because it has been discussed that corrupt monarchies are replaced by aristocracies and corrupt aristocracies are replaced by democracies. Option D is not the answer.
Choice (C)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 511

9. It must be understood that the author is against the speculations of critics of Wordsworth's the Ode.
Option A: The author believes a singular ***event*** cannot be the basis of a poet's faculty/ability. The author doesn't talk about a singular underlying ***emotion*** not being the basis of a poem. Hence, option A is not the answer.
Option B: This is to similar to what the author

7. Consider this: *'Both Sallust and Livy sought answers to the question, and if their explanations – the decline of morality ushered in by the very benefits of empire, with the concomitant loss of any considerable rival that could keep Rome in a state of readiness…'* Loss of rivals and decline of morality caused by the benefits of the empire/prosperity are two different factors to which Rome was exposed at the same time.

Option A: If false, it means Rome's prosperity didn't bring in more enemies. While this doesn't contradict Sallust and Livy (who believed Rome didn't have enemies during the phase of prosperity), it doesn't strengthen the argument either. Option A is not the answer.

Option B: If true, this strengthens the argument. So, if false, this option will weaken the argument. Option B is not the answer.

Option C: If we consider that Roman's moral strength didn't lay the foundation for its political prominence, we still don't have anything to evaluate Sallust and Livy's argument that moral decline was responsible for their political fall. Option C is not the answer.

Option D: If false, it means Romans didn't have to be in a state of readiness to protect the empire.

believes as understood from '*Nothing less than the whole mind, the whole man, will suffice for its origin…*' So, this will strengthen the author's argument. Option B is the answer.

Option C: The author says that Wordsworth's poetic faculty doesn't depend on one emotional circumstance. But we cannot say that his genius was not inspired by any emotional circumstance. The inspiration for his genius is different from the prerequisite for his genius. Critics, in the passage, imply that a single emotional circumstance is prerequisite for his genius, without which his poetic faculty fails. This option, which talks about the inspiration for his genius, does not find support either with the critics or with the author, since it is not relevant to the discussion. Hence, option C is not the answer.

Option D: While the author partially agrees with this, the author believes poetry is this and more. This option is closer to what critics of poetry feel. Option D is not the answer. Choice (B)

10. The author picks Dean Spenry's statement as a good representation of all the critics of Wordsworth's Ode. The author, however, doesn't agree with what the critics have to say.

Option A: The author's opinion about the statement is not that it has been 'ratified' or approved by other critics. The author singles out this statement to discuss as it was a good representation of the general critique of the Ode. Option A is not the answer.

Option B: This is not the author's opinion. Even though the author believes that the critics are not right when it comes to what they felt about the Ode, the author doesn't say that Wordsworth penned his farewell or anything of that sort. Option B is not the answer.

Option C: This is the author's opinion about Wordsworth's the Ode, but it was not offered as a criticism of the poem, but as a concession to why the ode is misunderstood/criticised. Option C is not the answer.

Option D: This represents the author's opinion about Spenry's statement and for that matter the opinion of all critics. The author doesn't agree with their attribution of Wordsworth's ability to write poetry to singular events. Option D is the answer. Choice (D)

Option B: The author does believe that the poem is dense (not perspicuous). However, that is not the argument the author was trying to make about the poem. The author believes that while it is true that the poem is dense, it has been misunderstood by critics. Option B is not the answer.

Option C: While this is what the author is arguing as a whole, this is not the purpose of mentioning the perspicuity of the poem itself. The author mentions that the poem is not perspicuous not to discuss the 'tacit assumptions' but to say that while the poem is hard to understand, that is not the reason why it has been misinterpreted. Option C is not the answer.

Option D: The wrong interpretation of the Ode, according to the author, stems from the inherent assumptions about poetry that people make, not from the lack of clarity in the poem itself or its being dense. The author accepts that the poem is hard to understand but believes that the poem is misinterpreted for other reasons. Option D is the answer. Choice (D)

11. The 'tacit assumptions' can be understood from the statements: '*The erroneous interpretation arises from certain extraneous and unexpressed assumptions… Nowadays it is not difficult for us to understand that* ***such tacit assumptions about the mental processes … I refer to the belief that poetry is made by means of a particular poetic faculty****, a faculty which may be isolated and defined.*'
Option A: If true, it is the equivalent of the tacit assumption, that the poetic faculty can be isolated and defined as a singular unit. If false, it will contradict the tacit assumption. Option A is the answer.
Option B: This, if true, contradicts the tacit assumption by stating that the poetic faculty is intangible and cannot be isolated or defined through one channel, Option B is not the answer.
Option C: Irrespective of whether this statement is true or false, it is irrelevant to what the author refers to as 'tacit assumptions' made by critics of the poetry. Option C is not the answer.
Option D: The tacit assumptions are ***about*** the mental processes (hidden, hence tacit), not that the mental processes are hidden beneath what we say. So, this option can neither agree with nor contradict the author as it is a misrepresentation of the discussion. Option D is not the answer.
Choice (A)

12. The author saying that the Ode is not perspicuous (perspicuous means clearly expressed or presented) is not a criticism of the Ode but a concession, before he goes on to defend the Ode against what critics had to say.
Option A: This is not the point the author is trying to drive home, considering the author was not critical of the poem, but of the critics. Option A is not the answer.

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 532

13. Option A: The study was not about the causes and effects of caloric restrictions. Rather, it was about how some traits are passed down generations even if not genetic. Hence, Option A is not the answer.
Option B: Inheritance of all traits doesn't happen at a molecular level without involving DNA. Genetic changes do involve DNA. Only epigenetics causes traits outside DNA. Hence, Option B is not the answer.
Option C: This can be understood from the study because the traits were not genetic and were not passed down through DNA but through epigenetics, as understood from '*Like the first generation of those caught up in the famine, the second generation also gave birth to low-weight children,* ***demonstrating the inheritance of traits at the cellular and molecular level operating apart from DNA.***' Option C is the answer.
Option D: We do not have sufficient information to establish that all second-generation traits are necessarily epigenetic. Also, the term second generation traits itself is ambiguous and doesn't have a fixed definition. Option D is not the answer.
Choice (C)

14. It can be understood from '***Evolutionary scientists refer to development as 'ontogeny':*** *even identical twins with the same DNA are different, epigenetically, because each person is embodied, embedded in relations or situations, in distinct and unpredictable ways.*'
Option A: Ontogeny is development, and not longing. Hence, Option A is not the answer.

Option B: This explains ontogeny or the evolution/development of an individual with genetic and epigenetic traits (hence, unique). Option B is the answer.
Option C: The author says that ontogeny is relevant both to scientists as well as to existentialists. However, ontogeny does not mean becoming both at the same time. Option C is not the answer.
Option D: Ontogeny doesn't refer to similarities and dissimilarities between two individuals. It is the development of an individual as a function of genetics and epigenetics. Option D is not the answer. Choice (B)

15. Presuppositions are tacitly accepted assumptions prior to further arguments. The author mentions examples of these presuppositions further: '*Are the fossils that indicate bipedal, two-legged movement the sign of human-hood? Or are signs of meaning-making behaviour, such as cave art or burying one's dead, better expressions of human animality because they point to symbolic, complex communication?*' The assumptions referred to here are the assumptions about what makes humans humans, and by extension different from other species. What defines human animality or human-hood.
Option A: If true, this helps us determine what makes humans special thereby answering our question. We can evaluate the importance of the artefacts left behind by our ancestors with respect to basic survival and evaluate the degree of human-hood based on this evaluation. Option A is not the answer.
Option B: If true, this helps us evaluate the presuppositions and come to a conclusion about what makes humans special. By studying the evolutionary changes and by checking at what point we diverged from the common ancestors, we can check at what point human-hood evolved. Hence, Option B is not the answer.
Option C: Even if true, this doesn't help us evaluate what makes humans special or different from other species they share a lot of common traits with. Option C is the answer.
Option D: If true, this explains or points out to what can be a separating trait between humans and other species/primates and what makes human-hood more prominent, or defines human animality. Option D is not the answer.

Choice (C)

16. Option A: If false, this option will show that modern humans have indeed descended from Homo Erectus, in which case it will agree with the author, and not contradict him. Option A is not the answer.
Option B: If false, this option indicates that modern humans are as similar to chimps and gorillas as they are to Homo Erectus, if not more. In that case, it doesn't necessarily contradict the statement that our emergence is intertwined with the story of the Homo Erectus. That can still be true. Option B is not the answer.
Option C: If proven to be false, this will indicate that hominisation didn't start with creation of fire, but with some other event. Irrespective of what that event is, it won't contradict the author's statement about the interconnectedness of Homo Erectus and modern humans. Option C is not the answer.
Option D: If false, this contradicts the author's argument that the history of modern humans is intertwined with the history of Homo Erectus. Option D is the answer. Choice (D)

Solutions for questions 17 and 18:

17. Option A: Our better understanding of hominins ('*Early hominins have been reported to be associated with densely wooded environments and sometimes forest*') has influenced hypothesis on what savannahs could be, and how we now think that the savannahs didn't play as significant a role in human evolutionary processes as was thought earlier. Option A is the answer.
Option B: This option establishes an incorrect causation: that savannah research has given us more clarity about what played a significant role in human evolution. It is early hominin studies that came up with such a thesis. Also, the option doesn't explain what the recalibration was. Option B is not the answer.
Option C: Savannahs are being seen as seasonal mosaic environments, based on compelling evidence. However, that they were seasonal mosaic environments didn't provide compelling evidence for their reduced role in human evolution. Option C is not the answer.
Option D: The role was not misunderstood, per se. It was the significance of the role that was not accurate. That has been clarified because of both hominin studies and our understanding of the savannahs. Option D is not the answer.
Choice (A)

18. Option A: This summary mentions all the valid points that: 1. Capitalism challenges inequality caused by social and cultural conventions and stratifications and 2. Capitalism has been misunderstood thanks to Marxist thought. Option A is the answer.
Option B: Capitalism doesn't allow skill and education to 'replace' social and cultural relationships. It replaces privilege and attacks these conventional relationships. This option also leaves out couple of important sub-ideas like the misrepresentation of capitalist societies by Marxist thought. Option B is not the answer.
Option C: The para doesn't say that liberal societies and capital societies are similar but separate. Also, this option misses out on important sub-ideas like what are the conventional sources of inequality. Option C is not the answer.
Option D: While this option is broadly true, it doesn't explain what the status quo is. According to the para, the status quo is perpetuated by social and cultural norms that cause inequality. Option D is not the answer. Choice (A)

Solution for question 19:

19. Sentence 4 starts with a connecting expression, 'In fact' which means it has to follow a sentence that says something slightly contrasting. 'Them' obviously refers to either meteorites or 'stones falling from the sky'. However, 4 talks about scientific community not paying them much heed, thereby, justifying the 'in fact' to connect to Sentence 1 which talks about how stones falling from the sky is 'a bit unsettling.' So, 14 is a block. Similarly, 'even when' in Sentence 3 connects to Sentence 4 which introduces the time frame 'two hundred years ago' when scholars began to take notice of these meteorites.

Sentence 2 has to precede 1 and also be the first sentence of the para because it starts with an outside topic about how details of the violent processes that formed Earth are preserved in meteorites. This is an introduction to meteorites, and it helps us understand what the author means by 'stones falling from the sky.' Please note that 1 cannot come in front of 2 because the first part of 1 and the second part of 2 (both about meteorites) are separated with the discussion about 'details of violent processes.' Therefore, 2143. Ans: (2143)

Solution for question 20:

20. Option A: This option leaves out the second part of the idea which is that the interpretation of facts by historians also subtly changes because of the facts. There is a reciprocation between the two processes - selection and interpretation. Option A is not the answer.

Option B: This option doesn't clarify as to how the reciprocity is obvious. In other words, it leaves out the influence of interpretation on selection of facts and vice-versa. Option B is not the answer.

Option C: This option summarises the para aptly by highlighting that the two processes, selection of facts by the historian and interpretation of those facts, affect each other, back and forth. This reciprocity between the past (facts) and present (historian's interpretation) is what the para is highlighting. Option C is the answer.

Option D: How a historian selects the facts influences the former's interpretation of those said facts, and that in turn influences the selection of

conventions have become too confining'. 2 explains why it has become confining. That brings us to the question of whether the '32' block comes first or the '14'. If '14' comes before '32' then in the sequence 1432, 3 cannot come after 4 because 4 talks about how scientific writing has changed (i.e., the situation in the present as compared to the past) and 3 again jumps back to the past ('over the past 50 years or so'). Hence, 1432 cannot be a valid sequence. Further, 3 is the bigger idea because it talks about 'scientists allowing the conventions of expression' whereas 4 talks about 'the stultifying convention'. Hence, 3214.

Ans: (3214)

Solution for question 22:

22. The given sentence talks about a complex symphonic work, as an analogy to talk about the genome and the instructions it carries. It can't be Option 1 because genome is only introduced in the next sentence. It can't be Option 2 either because 'the active blueprint' mentioned in the sentence after the blank 2 is a bigger idea, that has to precede any discussion of genome carrying instructions. Similarly, it cannot be Option 4 because the use of 'But' in the preceding sentence changes the tone, and is opposite to the given sentence. Also, 'score' is a synonym of symphonic work. Hence, Option 3 is the answer.

Choice (C)

Solution for question 23:

23. Sentence 1 starts with 'nonetheless' and has a negative tone towards national members of cartels. So, it would be preceded by the opposite, in Sentence 3 which talks about the clout of multinational corporations and market economies. Similarly, Sentence 1 will be followed by something on the same lines, Sentence 4, which starts with 'So long as' and elaborates on the tone of Sentence 1. Sentence 2 comes ahead of the 314 block because it introduces the dynamics between cartel nations and transnational corporations. Also, 3 continues to extend the second half of Sentence 2, by discussing those transnational corporations. Hence, 2314.

Ans: (2314)

facts. That is reciprocity. How we view the past, as per the para, is similar to the historian's interpretation of the past. Option D is not the answer.

Choice (C)

Solution for question 21:

21. The negative connector 'But' in 4 has to be preceded by something that is the opposite of 'art of writing suffered a grave setback'. So, 1 comes before 4. Similarly, 2 follows 3 because 2 is an elaboration of 3, where the author says 'the

Solution for question 24:

24. It cannot be Option 1 because the subsequent sentence is still talking about 'this condition' which refers to the first sentence of the given paragraph. The sentence fits perfectly in 2 because the subsequent sentence talks about 'these changes' referring to the ideas in the given sentence about social and economic change. Also, the idea of the given sentence—that it extends political consciousness, multiplies political demands, broaden political participation—makes perfect

sense when followed by how these changes undermine/weaken traditional sources of political authority. For the same reason, it cannot be Option 3. It won't be 4 because by then we have moved to the consequences of broadening political awareness and participation. Hence, Option 2 is the answer.

Choice (B)

Difficulty level wise summary - Section I

Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	8, 13, 14, 17, 19, 22, 24
Difficult	3, 4, 5, 6, 7, 9, 10, 12, 15, 18, 20, 21, 23
Very Difficult	1, 2, 11, 16

SECTION – II

Solutions for questions 1 to 5:

Let a be the number of Botany professors sent by A.
Total number of professors sent by A = a + 4 + 3 + 1 + 6 = a + 14
From (ii), this is one more than the total number of Botany professors sent by all the six countries.
The number of Botany professors sent by A, C and E is a + 8.
Hence, B, D and F must have sent a total of 7 professors.
From (i), the three countries must have sent 1, 2 and 4 Botany professors in any order (this is the only way in which the number of Botany professors will be distinct).
B could not have sent 2 or 4 Botany professors (as B has sent 2 Cosmology professors and 4 Psychology professors).
Hence, B must have sent 1 Botany professor.
Similarly, D could not have sent 4 Botany professors. Hence, D must have sent 2 Botany professors and F must have sent 4 Botany professors.

Let b be the total number of Zoology professors sent by B. The total number of professors sent by B must be b + 1 + 9 + 4 + 2 = b + 16
From (iii), the total number of Zoology professors is also b + 16 and the total number of Psychology professors is also b + 16.
From (iv), the total number of professors sent by C must be b + 27.
From (v), the total number of Cosmology professors sent by the six countries must be 2b + 32.
Since A, B, D, E and F sent a total of 21 Cosmology professors, the number of Cosmology professors sent by C must be 2b + 11.
The sum of Botany and Cosmology professors sent by C is 2b + 11 + 5 = 2b + 16.
Also, the total number of professors sent by C is b + 27.
Hence, the sum of Zoology, Sociology and Psychology professors sent by C must be 11 – b.
Since b is the number of Zoology professors sent by B, it cannot be 1 or 2 or 3 or 4. Hence, the minimum value of b is 5.

If b is 5, then the sum of Zoology, Sociology and Psychology professors sent by C must be 6.
Since the number of professors sent by C must be distinct, this sum cannot be less than 6 (since 1 + 2 + 3 = 6) and hence, b cannot be more than 5.
The number of Zoology, Sociology and Psychology professors sent by C must be 1, 2 and 3, in any order, and b must be 5.
C cannot have sent 1 or 2 Psychology professors (since A and E sent those many Psychology professors).
Hence, C must have sent 3 Psychology professors. The number of Zoology and Sociology professors sent by C can be 1 or 2.
Since C sent b + 27 = 32 professors in total, C must have sent 21 Cosmology professors.
Since b = 5, we can fill all the values which we have expressed in terms of b.

The following table provides this information:

Country	Botany	Zoology	Sociology	Psychology	Cosmology	Total
A	a	4	3	1	6	a + 14
B	1	5	9	4	2	21
C	5	1/2	2/1	3	21	32
D	2	3	5		4	
E	3			2	8	
F	4	6			1	
Total	a + 15	21		21	42	

If C sent 1 Zoology professor, for the total number of Zoology professors to be 21, E must have sent 2 Zoology professors. However, this is not possible as E has sent 2 Psychology professors.
Hence, C must have sent 2 Zoology professors and E must have sent 1 Zoology professor.
The total number of Psychology professors sent by the six countries is 21. The number of Psychology professors sent by A, B, C and E combined is 10. Hence, D and F must have sent 11 Psychology professors in total.
Since D and F could not have sent 1 or 2 or 3 or 4 Psychology professors, D and F must have sent 5 and 6 Psychology professors, in any order.
Since D sent 5 Sociology professors, D must have sent 6 Psychology professors and F must have sent 5 Psychology professors.
From (v), F must have sent 35 professors in total. Hence, F must have sent 19 Sociology professors.
From (vi), A must have sent 29 professors in total. Hence, a must be 15.
From (vi), the number of Sociology professors sent by E must also be 15.
The following table provides the number of professors sent by each country:

Country	Botany	Zoology	Sociology	Psychology	Cosmology	Total
A	15	4	3	1	6	29
B	1	5	9	4	2	21
C	5	2	1	3	21	32
D	2	3	5	6	4	20
E	3	1	15	2	8	29
F	4	6	19	5	1	35
Total	30	21	52	21	42	166

1. C sent 21 Cosmology professors. Ans: (21)

2. E sent the least number of Zoology professors. Choice (B)

3. F sent the maximum total number of professors. Choice (A)

4. A sent 15 Botany professors. Ans: (15)

5. 52 Sociology professors were sent by the six countries combined. Choice (B)

Solutions for questions 6 to 10:

Given that the horizontal axis represents the average number of customers per day visiting any store and the area of the rectangle represents the revenue of the store.
Since the horizontal axis multiplies by 365 provides the total number of customers, the width of any rectangle is proportional to the number of customers who visited the store.
Since the area is the total revenue, the height of the rectangle will be proportional to the average revenue per customer. Only then will the product of the height and width provide the revenue of the store.
We can observe that the highest ARPC is for the orange rectangle with revenue of 800. Hence, from (ii), ***orange represents South***.

From (i), the rectangle representing the store with the least revenue in West must have the same height as one of the other rectangles, which will represent the store in the North.
If West is represented by grey, then the least revenue is 99. No other rectangle has the same height as the rectangle which represents revenue of 99. Hence, this is not possible.
If West is blue, then the least revenue is 252. Again, no other rectangle has the same height as this rectangle.
Hence, ***West must be green***. The least revenue is 144. The height of this rectangle is the same as the grey rectangle with revenue of 144. Also, for grey, this rectangle represents the second highest value. Hence, ***grey must represent North*** and therefore, ***blue must represent East***.

6. The highest revenue of any store in the North is 900. Ans: (900)

7. The least ARPC of a store is the store in North with revenue of 99 (since this has the least height).
The total number of customers for this store = 4.4 × 365 = 1606.
The ARPC of this store = 99000/1606 = 61.6
Hence, for a store to have ARPC greater than 240, it must be approximately 4 times the height of this rectangle.
We can check for the store in South with a revenue of 63.
The ARPC for this store = $\frac{63000}{0.8 \times 365}$ = 215.75
Since this is not satisfying the given condition, we can check for any of the two stores with revenue of 144.

The ARPC of these stores = $\frac{144000}{1.6 \times 365}$ = 246.57
Hence, these stores satisfy the given condition. Also, all the stores which are represented by rectangles with a greater height will also satisfy the condition.
By observation, we can see that there are 10 rectangles with a greater height.
Hence, 10 stores satisfy the given condition. Ans: (10)

8. The total number of customers who visited any store in the East = (4 + 2.8 + 1.6) × 365 = 3066
The total number of customers who visited any

9. The highest ARPC (i.e., the rectangle with the maximum height) is the store in South with revenue of 800.
The ARPC of the store = $\frac{800000}{3.4 \times 365}$ = 644.64 ≈ 645 Choice (B)

10. To find the region in which the total number of customers who visited any store in that region the highest, we only have to find the sum of the average number of customers for the stores in each region.
For West, the sum is 3.4 + 2 + 3.2 + 1.6 = 10.2
For East, the sum is 4 + 2.8 + 1.6 = 8.4

store in the North = $(4 + 4.4 + 1.6) \times 365 = 3650$
Required difference = $3650 - 3066 = 584$.
Choice (D)

For North, the sum is $4 + 4.4 + 1.6 = 10$
For South, the sum is $3.4 + 4 + 0.8 + 0.8 = 9$
Hence, the total number of customers who visited any store in that region is the highest for West.
Choice (B)

Solutions for questions 11 to 15:

Given that the three persons made the statements one after the other.
From the way the three persons made their statements, the first statement cannot be made by A (since the number of truths already made should be at least one). Also, A could have made only the odd numbered statements (only then can the number of truths and lies be equal).
Also, the first statement could not have been made by B, since the number of true statements made before this was not an odd number.
Hence, the first statement must have been made by C (since the false statements made before this was 0, an even number).

The second statement cannot be made by C (because he made the previous statement) and it cannot be made by A (since the number of truths and lies cannot be same). Hence, the second statement must have been made by B. For the second statement to be made by B, the first statement must be true.

The third statement can be made by A or C.
If the third statement was made by C, then the second statement must be true. The fourth statement must have been made by B. For B to make the fourth statement, the third statement must be true.
Hence, the first three statement will be true in this case. Then A cannot make any statements until there are at least three false statements (making it a total of six statements). In this case, A can only make statements 7, 8 and 9, which is not possible. Hence, the third statement could not have been made by C.
∴ A made the third statement. Since A made the third statement, the second statement must be false.
The fourth statement must be made by B or C. Irrespective of who makes the fourth statement, the third statement must be false.
If the fourth statement is false, then fifth and sixth statements must be true and A must have made the 7th and 9th statements. The fifth statement must have been made by B (since there was only 1 true statement before this). The sixth statement could not have been made by A or B or C. Hence, the fourth statement cannot be false.
The fourth statement must be true.

Now, looking at the statements, we know that C's first statement is true. Hence, Kishore has three kids.
We also know that the second statement, i.e., B's first statement, is false.
Hence, Lalit lives in Bangalore.
The third statement, i.e., A's first statement, is false. Hence, the person with two kids does not live in Hyderabad.
The fourth statement can be made by B or C, but it has to be true. B's second statement, i.e., Naveen has three kids, is false because Kishore has three kids. Hence, B could not have made the fourth statement. C must have made the fourth statement.
Since this is true, Amit does not live in Chennai and has more than one kid.
The fifth statement could not have been made by C or B. Hence, A must have made the fifth statement.
The sixth statement must have been made by B or C. Irrespective of who made this statement, the fifth statement must be true.
Hence, the person with three kids must live in Mumbai. Since Kishore has three kids, Kishore must be in Mumbai. Since Lalit lives in Bangalore and Amit does not live in Chennai, Amit must be in Hyderabad. Since the person from Hyderabad does not have 2 kids, Amit must have 4 kids. Hence, Naveen lives in Chennai and has 1 or 2 kids. Lalit also has 1 or 2 kids.

If C made the sixth statement, then among statements 7, 8 and 9, B must make two statements and A must make 1. A cannot make statement 8 (since A can only make odd numbered statements). In this case, B has to make two consecutive statements, which is not possible.
Hence, B must have made the sixth statement, which is B's second statement. Since B's second statement is false, the sixth statement must be false.
The seventh statement cannot be made by C, since the lies made are not even. Hence, A must have made the seventh statement.
Irrespective of who made the eighth statement, the seventh statement must be false.
Hence, the seventh statement, i.e., A's third statement, is false. The person with one kid does not live in Bangalore. Hence, Lalit must have 2 kids and Naveen must have 1 kid. We now know the distribution and it is given below:

Person	City	Kids
Amit	Hyderabad	4
Kishore	Mumbai	3
Lalit	Bangalore	2
Naveen	Chennai	1

The eighth and ninth statements must have been made by B and C. Also, the two statements are true. If B made the ninth statement, then the eighth statement made by C must be false, which is not possible. Hence, C made the ninth statement and B made the eighth statement. Both of these are true.
The following table provides the persons who made each statement and whether each statement is true or false:

Statement	1	2	3	4	5	6	7	8	9
True/False	T	F	F	T	T	F	F	T	T
Person	C	B	A	C	A	B	A	B	C

11. Kishore lives in Mumbai. Choice (C)

12. The three persons made five true statements. Ans: (5)

13. Everyone made at least one true statement. Choice (A)

14. Naveen has 1 kid. Choice (C)

15. Only Statement 7 is false. Hence, the answer is option A. Choice (A)

Solutions for questions 16 to 20:

From (ii), A and B were the Prosecution and Defence Lawyers, respectively, for 11:20 am case.
From (v), the cases scheduled at 10:00 am, 11:20 am and 4:20 pm were won by the Defence Lawyers. Hence, B must have won the 11:20 am case.
From (v), these cases cannot be C2 or C4. From (ii), the case with A and B cannot be C1 or C5. Hence, it has to be C3 or C6. However, from (vi), it cannot be C3, since G was a lawyer for C3. Hence, the 11:20 am case must be C6.
From (i), F had two cases after 12 noon. Since there must be at least 120 minutes between any two cases, F's cases must be at 1:30 pm and 4:20 pm.
From (iii), C had three cases. He could not have had the 11:20 am case because A and B are the lawyers for that case. If he had the 11:50 am case, then C cannot have three cases. Hence, C must have the cases at 10:00 am, 1:30 pm and 4:20 pm.
The case at 11:50 am is C3 (from (vi)) and G is one of its lawyers.
Since E won at least one case (from (vi)) and he lost the only case he was a Defence lawyer for, E must have won a case as the Prosecution lawyer. Therefore, 3 cases were won by the Prosecution and 3 cases were won by the Defence.

Given that C4 was won by the Prosecution. The cases at 10:00 am and 4:20 pm cannot be C4, since these cases were won by the Defence. C4 could have been the case at 1:30 pm or 2:40 pm. If C4 was the case at 2:40 pm, it must have been won by E and C2 must be the case at 1:30 pm. But the case at 1:30 pm had C and F as lawyers. Therefore, C4 could not have been the case at 2:40 pm. Hence, the case at 1:30 pm must be C4, which was won by the Prosecution.
From (iv), for C2, E is the Defence Lawyer and D is the Prosecution Lawyer. The only case that can be C2 is the 2:40 pm case. This case must have been won by D.
The case at 10:00 am and 4:20 pm must be C1 or C5 in any order.

Since E was the Defence Lawyer for only one case, the cast at 10:00 am could not have been won by E (since this case was won by the Defence Lawyer). If E was a lawyer of this case and lost it, he could not have been the lawyer for C3, the only other case that E can win. Hence, E could not have won any cases in this scenario, which violates (vi). Hence, E must have won C3 against G. Since E cannot be the Defence Lawyer, E must be the Prosecution.
From (i), D was the Prosecution for two cases. Hence, D must be the Prosecution for the 10:00 am case (along with C2, the 2:40 pm case). C must have won this case as Defence.
From (iii), C won a case as Prosecution. Hence, C must have won C4, the 1:30 pm case, against F. Since C won only two cases, C must have lost the 4:20 pm case to F, who is the Defence Lawyer for that case.

The following table provides the details of the six cases:

Time	Case	Prosecution	Defence	Won By
10:00 am	C1/C5	D	C	C
11:20 am	C6	A	B	B
11:50 am	C3	E	G	E
1:30 pm	C4	C	F	C
2:40 pm	C2	D	E	D
4:20 pm	C5/C1	C	F	F

16. Only one lawyer, B, did not lose any case on that day. Choice (C)

17. C was the Defence Lawyer for either C1 or C5. Choice (D)

18. D lost a case only against C. Choice (A)

19. E was the Prosecution Lawyer for only C3. Choice (B)

20. C4 was won by C. Choice (C)

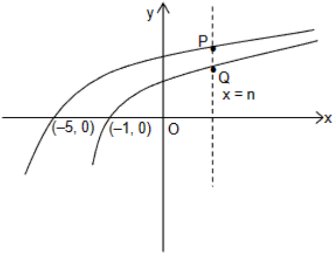
Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	7, 9
Medium	6, 8, 10, 16, 17, 18, 19, 20
Difficult	1, 2, 3, 4, 5
Very Difficult	11, 12, 13, 14, 15

SECTION – III

Solutions for questions 1 to 6:

Now, if A completes $\frac{1}{5}^{th}$ of the work, and B does the remaining $\frac{4}{5}^{th}$, we get $\frac{1}{5} \times 30 + \frac{4}{5} \times 20 = 6+16$ = 22 days. Ans: (22)

2. The graph of the curves will look as given below:
The given distance between points P and Q is simply the difference between the y-coordinates at *x* = *n*.



1. Let the number of days taken by A alone and B alone be a and b respectively. Since each of A and B worked on exactly half the work and finished it in 25 days

$\frac{a}{2} + \frac{b}{2} = 25 \Rightarrow a + b = 50$ ____ (1)

Also, together they take 12 days.

$\Rightarrow \frac{ab}{a+b} = 12$ _____ (2)

Solving (1) and (2), $a = 30$, $b = 20$ (since A is slower)

It is given that $\log_9(n + 6) – \log_9(n + 2) = \frac{1}{4}$

$\Rightarrow \log_9 \frac{(n+6)}{(n+2)} - \frac{1}{4}$

$\Rightarrow \frac{n+6}{n+2} = (9)^{\left(\frac{1}{4}\right)}$

$\Rightarrow n + 6 = \sqrt{3}\,(n + 2)$

$\Rightarrow (\sqrt{3} – 1)\,n = 6 – 2\sqrt{3}$

$\Rightarrow n = 2\sqrt{3}$

$\therefore \frac{n}{\sqrt{3}} = 2$ Ans: (2)

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3. Let R be the radius of the quadrant.

Area of the quadrant = $\frac{\pi}{4}(R^2) = 154$

$\Rightarrow R = 14$

The perimeter of the quadrant = $\frac{2\pi R}{4} + 2R$

$= \frac{2 \times \frac{22}{7} \times 14}{4} + 2(14) = 22 + 28 = 50$ cm

Choice (C)

4. Let the five integers be denoted by a, b, c and e respectively.

It is given that $\frac{a+b+c+d}{4} + e = 41$ ----- (1)

$\frac{a+b+c+e}{4} + d = 44$ ----- (2)

$\frac{a+b+d+e}{4} + c = 50$ ----- (3)

$\frac{a+c+d+e}{4} + b = 56$ ----- (4)

$\frac{b+c+d+e}{4} + a = 65$ ----- (5)

Adding the five equations given above, we get $2(a + b + c + d + e) = 256$

$\Rightarrow a + b + c + d + e = 128$

Substituting $128 – e$ for $a + b + c + d$ in equation (1), we get $\frac{(128 - e)}{4} + e = 41 \Rightarrow e = \frac{(41) \times 4 - 128}{3}$

$\Rightarrow e = 12$

6. $675 = 3^3\,5^2$. There are 12 factors and 6 ways of expressing 675 as a product of 2 factors. One of the factors is not a multiple of 3, i.e., it is $3^0\,5^x$. One of the factors in not a multiple of 5. There are 2 cases. The non-multiple of 5 is the same as the non-multiple of 3, i.e., the factor is $3^0 5^0$ (The other factor would be $3^3 5^2$). Alternately, the non-multiple of 5 can be different from the non-multiple of 3, i.e., the factors are 5^2 (the non-multiple of 3) and 3^3 (the non-multiple of 5). Thus, there are only two rectangles which satisfy the given conditions.

Choice (A)

Solutions for questions 7 and 8:

Since $abcde$ is a five-digit number with $a < b < c < d < e$ and divisible by 6, e must be 6 or 8.

7. Given $e < 8 \Rightarrow e = 6$;
Then $a + b + c + d$ must be a multiple of 3 and each digit is less than or equal to 5.
∴ The only possible value of $abcd$ is 1245.
(The five-digit number = 12456) Choice (A)

8. In addition to the possibility of $e = 6$, we need to consider the case when $e = 8$.
If $e = 8$, then $a + b + c + d$ must be of the form $3k + 1$ and each digit is less than or equal to 7.
For the digits 1, 2, 3, 4, 5, 6, 7, the remainders when divided by 3 are 1, 2, 0, 1, 2, 0, 1 respectively. The sum 1 + 2 + 3 + 4 + 5 + 6 + 7 = 28 is itself of the form $3k + 1$. Hence exactly three of these seven digits, with their sum adding to a multiple of 3, need to be removed (and the

Similarly, we get $d = \frac{(44)\times 4-128}{3} = 16$

and $c = \frac{(50)\times 4-128}{3} = 24$

and $d = \frac{(56)\times 4-128}{3} = 32$

and $a = \frac{(65)\times 4-128}{3} = 44$

Among the choices given, one of the integers is 44.

Choice (B)

5. From the given information it is clear that for the n^{th} G.P the first term is $n + 1$ and the common ratio is $\frac{1}{n+2}$

$\therefore S_n = \frac{n+1}{1-\frac{1}{n+2}} = n + 2$

Therefore, $S_1 + S_2 + \text{--------} + S_n = \Sigma (n + 2)$

$\therefore \sum_1^{20} S_n = [3 + 4 + 5 + ...22] = \frac{(3+22)}{2} \times 20 = 250$

Choice (D)

remaining four will form *abcd*).

Now the remainders of the three digits removed need to add up to a multiple of 3. This is possible in two ways.

(i) Remainders of 1, 2 and 0 are chosen.
(ii) Remainders of 1, 1 and 1 are chosen.

Since there are three 1's, two 2's and two 0's, there are 3 × 2 × 2 = 12 combinations in (i) and since there are only three 1's, there is only one combination in (ii)

Hence, the possible number of values of *abcd*, if *e* = 8 is 12 + 1 = 13.
Therefore, the total number of possible values of *abcde* = 1 + 13 = 14. Choice (B)

Solutions for questions 9 to 22:

9. Let $f(x) = ax^2 + bx + c$
It is given that $f(4) = 4f(1)$
$\Rightarrow 16a + 4b + c = 4(a + b + c)$
$\Rightarrow c = 4a$ ----- (1)
Further, since 3 is one of the roots,
$f(3) = 0$
$\Rightarrow 9a + 3b + c = 0$
Putting $c = 4a$, (from (1)),
we get $9a + 3b + 4a = 0$

$b = -\frac{13}{3}a$

The quadratic equation is therefore
$ax^2 + bx + c = 0$
i.e., $ax^2 - \frac{13}{3}ax + 4a = 0$
$\Rightarrow a(3x^2 - 13x + 12) = 0$
$\Rightarrow a(3x - 4)(x - 3) = 0.$

Hence, the other root $\alpha = \frac{4}{3}$,

and $12\propto = 12 \times \frac{4}{3} = 16.$

Alternative Solution 1:

Once we arrive at $c = 4a$ (i.e., eq (1) in above solution), we can see that $\frac{c}{a} = 4$ (i.e., product of roots $(\alpha \times \beta) = 4$). Hence, $3\alpha = 4 \Rightarrow 12\alpha = 16$.

13. The area of a parallelogram is bh, where b is the length of the base and h is the height of the parallelogram.
We can consider any one side as the base of the parallelogram and we can find the corresponding height (the distance between parallel sides).
Let us consider, 2x + 3y – 5 = 0 as the base.
Then, the vertices of the base are the intersection points of
(2x + 3y – 5 = 0, 3x + 2y – 5 = 0) and
(2x + 3y – 5 =0, 3x + 2y – 10 =0), i.e., (1, 1), (4, –1).

$\therefore$ The length of the base (b) = $\sqrt{(1-4)^2+(1-(-1))^2}$

$= \sqrt{3^2+2^2} = \sqrt{13}$ units.

The height of the parallelogram is the distance between

2x + 3y – 5 = 0 and 2x 3y – 10 = $\frac{|c_1 - c_2|}{\sqrt{a^2+b^2}}$

Alternative Solution 2:

Let the function be expressed as $f(x) = k(x - \alpha)(x - 3)$.

Since $f(4) = 4f(1)$, $k(4 - \alpha)(4 - 3) = 4k(1 - \alpha)(1 - 3)$

$\Rightarrow 4k - k\alpha = 8k\alpha - 8k$

$\Rightarrow 9\alpha = 12$, i.e., $12\alpha = 16$. Ans: (16)

10. The values of the given function and the functions given in the choices are tabulated below for $2 \le x < 4$

	$x = 2$	$2 < x < 4$
$f(x)$	1	1
Statement I	1	2
Statement II	0	1
Statement III	1	1

We see that only statement III is true.

Choice (D)

11. It is given that,

The distance covered, $(d) \alpha t^2$.

$\therefore d = kt^2$.

Distance covered in first 10 seconds $= k(10)^2 = 300$

$\Rightarrow k = 3$.

Distance covered in the first 20 seconds $= k(20)^2 = 1200$

Therefore, the body covers 1200 – 300 = 900 m in the next 10 seconds. Ans: (900)

12. Consider 7^4, whose value is 2401

$\therefore 7^{700} = (7^4)^{175} = (2401)^{175}$

Any power of 2401 will end with 1 as the units digit and 0 as the tens digit.

$\therefore$ When it is divided by 100, the remainder is 1.

Ans: (1)

$$= \frac{|-5-(-10)|}{\sqrt{2^2+3^2}} = \frac{|5|}{\sqrt{13}} \text{ units}$$

$\therefore$ The area of the parallelogram = bh

$$= \sqrt{13}\left(\frac{5}{\sqrt{13}}\right) = 5 \text{ sq.units.}$$

Alternative Solution:

Let the parallelogram's vertices be A, B, C, D in that order:

By solving for the equations, we get A(1, 1), B(–1, 4), C(2, 2) and D(4, –1)

Lengths AB = BC = CD = DA

Hence, ABCD is a rhombus

$\therefore$ Area $= \frac{1}{2} AC \times BD$

$= \frac{1}{2} \times \sqrt{2} \times 5\sqrt{2}$

= 5 sq. units Ans: (5)

14. If $x = -8$ and $y = -6$, $x + y = -14$

Hence, choice (A) is false.

If $x = -8$ and $y = 6$, $xy = -48$.

Hence, Choice (B) is also false.

For $\frac{x}{y}$, the minimum value can be realised when $x = -8$ and $y = 1$, while the maximum value can be realised when $x = -8$ and $y = -1$.

That is $-8 \le \frac{x}{y} \le 8$.

Hence, Choice (C) is true. Choice (C)

15. Given $(K^2 - 10K + 21)(K^2 - 9K + 20) = 240$

$\Rightarrow (K - 3)(K - 7)(K - 4)(K - 5) = 240$

This is of the form $A(A + 2)(A + 3)(A + 4)$, where $(K - 7) = A$.

Now, factorising 240 as $2^4 \times 3 \times 5 = 2 \times 4 \times 5 \times 6$ we can see that, for K = 9, we get

$(K - 7)(K - 5)(K - 4)(K - 3) = 2 \times 4 \times 5 \times 6 = 240$

$\therefore$ The value of K is 9.

Since, K is a natural number, and since all four terms on the LHS are positive for $K \geq 9$, this is the only possible value of K (because for any value of $K > 9$, the product will be greater than 240).

Ans: (9)

16. Let the initial quantities of milk and water be $3x$ and x.

The volume of water added to increase the total volume by 50% will be $\frac{(3x+x)}{2} = 2x$.

Now, quantities of milk and water will be $3x$ and $(x + 2x)$, i.e., the ratio $= 3x : 3x$, i.e., 1 : 1. Hence, any volume of mixture removed will have equal quantities of milk and water. That is, 25 litres of mixture that was removed must have had 12.5 litres each of milk and water.

$\therefore$ Ratio of Final quantities after replacing 25 litres with water $= \frac{3x - 12.5}{(3x - 12.5) + 25} = \frac{3x-12.5}{3x+12.5} = \frac{1}{3}$ (given)

$\Rightarrow 3x = 25$

Now initial quantity of mixture was $3x + x = 4x$

$= \frac{25 \times 4}{3} = 33\frac{1}{3}$ litres.

Choice (B)

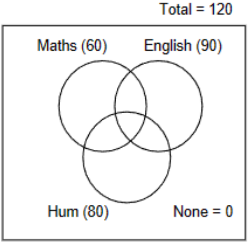
17. The first day since Rishabh and his mother reach home $13\frac{1}{3}$ minutes earlier than usual, his mother must have saved $6\frac{2}{3}$ minutes either way, i.e. she met Rishabh at $\left(4:00 - 6\frac{2}{3}\right)$ or at $3 : 53\frac{1}{3}$.

The distance that Rishabh covered in $53\frac{1}{3}$ minutes, his mother normally covers in $6\frac{2}{3}$ minutes. i.e., the mother is $\frac{53\frac{1}{3}}{6\frac{2}{3}}$ i.e., 8 times as fast as Rishabh.

The second day, school got over at 3:30 pm. Dividing 30 minutes (4 : 00 – 3 : 30) into 9 parts we get each part as $3\frac{1}{3}$ minutes. Rishabh uses up 8 of these parts to cover a certain distance, while his mother would need only 1 part to cover this distance. On that day, she did not cover this distance (as Rishabh had already covered it). She saved $3\frac{1}{3}$ minutes one way and $6\frac{2}{3}$ minutes on the to and fro trip.

Choice (C)

18. Let the number of students who passed in exactly one, exactly two and exactly three subjects be a, b and c respectively.



$a + b + c = 120$

$a + 2b + 3c = 60 + 90 + 80$

Subtracting $b + 2c = 110$ _______ (1)

It is given that, $a + b + c = \frac{5}{4}(b + c)$

$120 = \frac{5}{4}(b + c)$

$\therefore b + c = 96$ ________ (2)

Solving (1) and (2), we get $c = 14$. Choice (D)

19. $\frac{7}{b} + \frac{8}{a} = -1$

$\Rightarrow 7a + 8b = -ab \Rightarrow 7a + 8b + ab = 0$

$\Rightarrow (a + 8)(b + 7) - 56 = 0 \Rightarrow (a + 8)(b + 7) = 56$

$= 2^3(7)$

So, 56 has 4(2) or 8 positive factors and it can be expressed as the product of 2 factors in 4 ways. We need not consider negative factors as we want $a + b$ to be positive. Of these 4 ways, we need to look at the cases where $(a + 8) + (b + 7)$ is greater than 15.

$56 = 1(56) = 2(28) = 4(14) = 7(8)$

We ignore the last case and accept only the first 3 cases. This would correspond to 6 solutions (for example, $a + 8 = 1$, $b + 7 = 56$ and vice versa).

Choice (A)

20. $T = 2\cos^4\theta + \sin^2\theta + 3$

$T = 2\cos^4\theta + 1 - \cos^2\theta + 3$

$T = 2\cos^4\theta - \cos^2\theta + 4$

$T = 2[\cos^4\theta - \frac{1}{2}\cos^2\theta + 2]$

$T = 2[(\cos^2\theta - \frac{1}{4})^2 + 2 - \frac{1}{16}]$

$0 \leq \cos^2\theta \leq 1$

$\frac{-1}{4} \leq \cos^2\theta - \frac{1}{4} \leq \frac{3}{4}$

$0 \leq (\cos^2\theta - \frac{1}{4})^2 \leq \frac{9}{16}$

$2\left[2 - \frac{1}{16}\right] \leq T \leq 2\left[\frac{1}{16} + 2 - \frac{1}{16}\right]$

$\frac{31}{8} \leq T \leq 5$

$T \in \left[\frac{31}{8}, 5\right]$

Alternative Solution:

By inspection, it can be observed that for cosθ = 1, the expression can be equal to 5. Also, since $\cos^4\theta$ and $\sin^2\theta$ are both positive, the expression cannot be less than 3. By observing the choices, only choice (C) can be the answer. Choice (C)

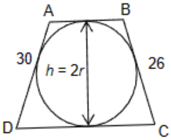
21. $123.01 = (1 \times 4^2 + 2 \times 4 + 3 \times 4^0) + \left(2 \cdot \frac{1}{4} + 1 \cdot \frac{1}{4^2}\right)$

$= (16 + 8 + 3) + \left(\frac{1}{2} + \frac{1}{16}\right) = 27 + (0.5 + 0.0625)$

$= 27 + 0.5625 = 27.5625$

Ans: (27.5625)

22. The size of the plot will be the minimum when all the four sides of the plot (i.e. trapezium) touch the pool.



Let the oblique sides be AD and BC. Since the circle is inscribed in ABCD, the sum of the lengths of opposite sides will be equal.

Hence, AB + DC = AD + BC = 30 + 26 = 56 cm

Also, height of the trapezium = $h = 2r$ = diameter of circle

$= 12 \times 2 = 24$ m

Hence, area of trapezium $= \frac{1}{2}$ (AB + DC) $\times h$

$= \frac{1}{2} \times 56 \times 24$

= 672 sq. m Choice (B)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	–
Easy	3, 11, 14
Medium	1, 5, 7, 9, 10, 12, 16, 18, 19, 21
Difficult	2, 4, 6, 8, 13, 15, 17, 20, 22
Very Difficult	–